

HOLLYWOOD

FIU Asset Management - 2025 Portfolio Report

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Foreword

ABOUT THE AUTHORS

Nguyen Hai Duy	2212345017
Pham Thanh Huyen	2212345023
Tran Ngoc Phuong Uyen	2212345051
Nguyen Tri Minh Thu	2212345043
Le Tran Quynh My	2212345028

() These authors contributed equally to this work and share first authorship.*

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Thank you for giving us the opportunity to work on this assignment, which has greatly enhanced our understanding of the subject and allowed us to apply theoretical knowledge to practical analysis.

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*(**) This section is excluded from the word count.*



List of Contents

ACTIVE FUND	4
1.EXECUTIVE SUMMARY	5
2.ASSET ALLOCATION AND SECURITY ANALYSIS.....	7
3.PORTFOLIO CONSTRUCTION AND REBALANCING	11
4.PERFORMANCE AND RISK EVALUATION	
PASSIVE FUND	17
1.EXECUTIVE SUMMARY.....	18
2.PORTFOLIO CONSTRUCTION AND REBALANCING	20
3.PERFORMANCE AND RISK EVALUATION	22
REFLECTION ANH LIMITATIONS	24
REFERENCE	26



ACTIVE FUND



EXECUTIVE SUMMARY

FUND DESCRIPTION

Our Investment Philosophy

We believe economic movements and sector cycles are key forces that shape market leadership and help indicate where capital is most effectively deployed. We also recognize that equity prices can drift away from fundamental value and often reflect behavioural patterns, creating opportunities for disciplined investors who maintain balanced exposure to both growth and value. Moreover, we believe that in a data-driven environment, combining a solid fundamental perspective with a validated quantitative approach helps overcome behavioural biases and supports stronger risk-return outcomes over time.

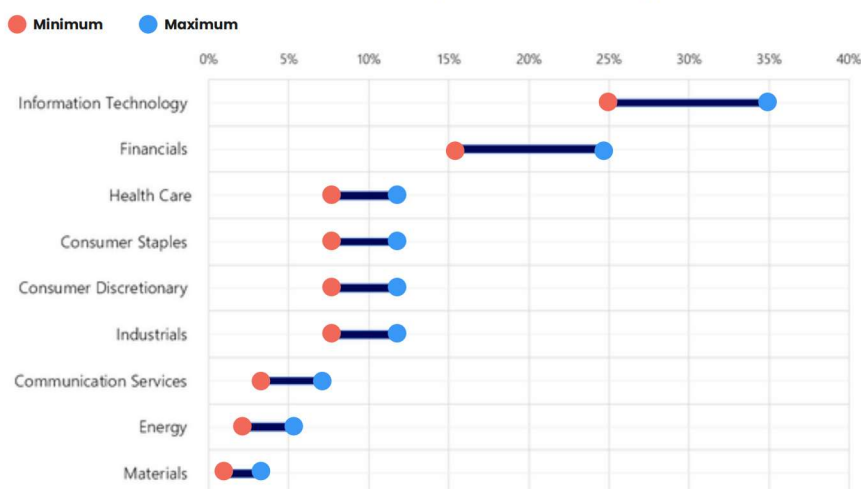
Our Investment Objective

- **Benchmark:** Dow Jones Industrial Average (DJIA)
- **Annualized Return Target:** DJIA + 10–15%
- **Expected Total Return:** 8–10% p.a. under normal market conditions
- **Volatility Target:** < 20% (annualized)
- **Sharpe Ratio Target:** ≥ 1.25
- **Maximum Drawdown Limit:** < 20%
- **Beta (β):** Dynamic; adjusted via futures (≈ 1.0 baseline)

Portfolio Breakdown

FIGURE 1 | How the portfolio is allowed to allocate across sectors

Minimum and maximum sector weights, investment guidelines



Fund Information

Inception Date	1 October, 2024
AUM	\$ 50 million
Benchmark Index	Dow Jones Industrial Average

Our Investment Style

- Large - cap • Core • Momentum
- Sector Rotation

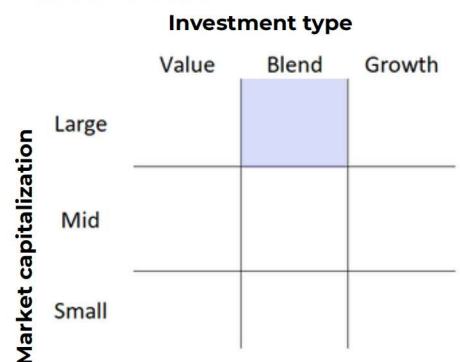
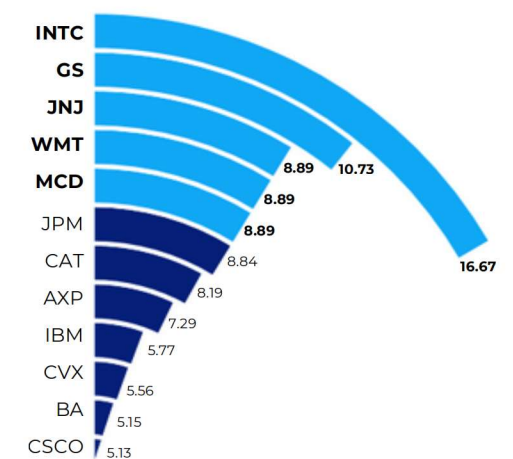


FIGURE 2 | Weight of the top 10 holdings

Portfolio Weight Distribution (%)

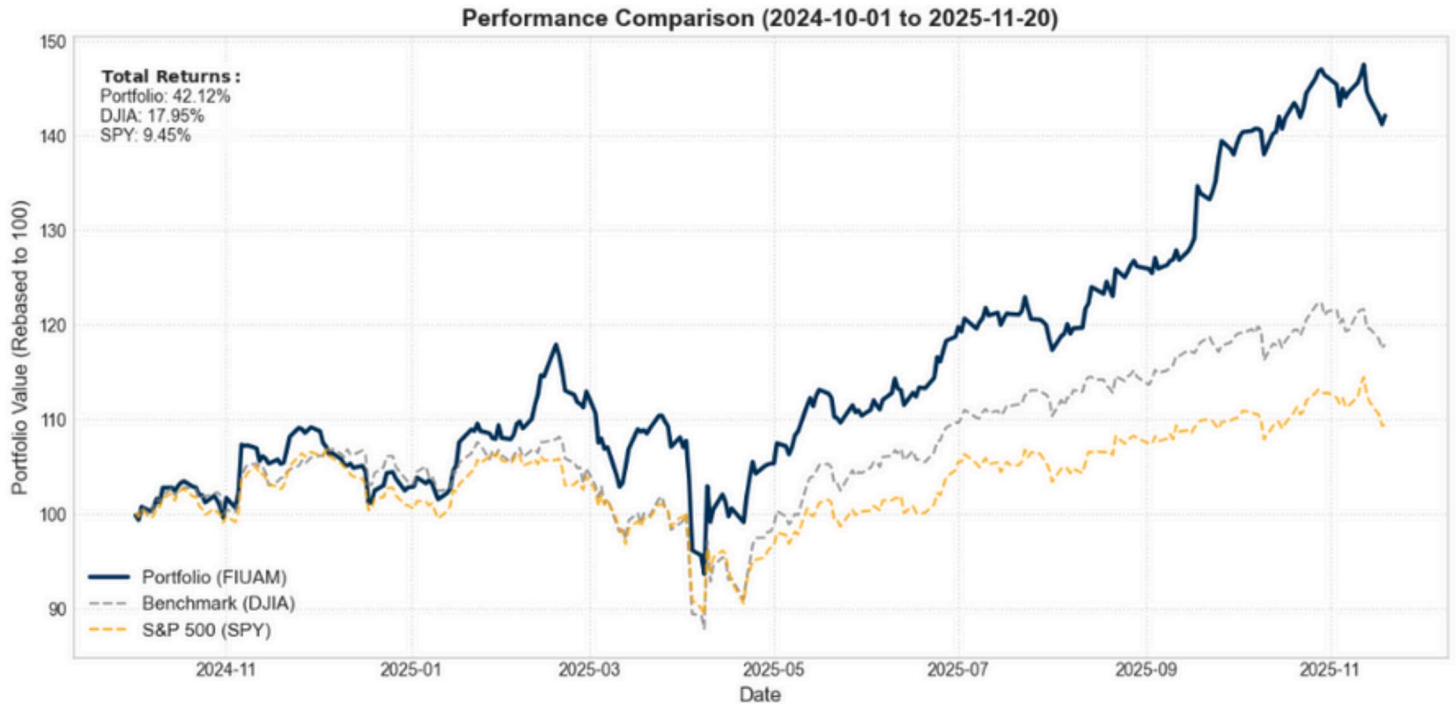


(*) Data derived from the group's Python dataset.

PERFORMANCE SUMMARY

FIGURE 3 | Beating the Benchmarks?

Cumulative returns of the portfolio vs. market benchmarks



(*) Chart generated using the group's Python code.

TABLE 1 | The portfolio consistently exceeds benchmarks in cumulative and risk-adjusted returns.

Detailed Performance Interpretation

	Total Return	Ann. Return (CAGR)	Ann. Volatility	Sharpe Ratio	Max Drawdown
Portfolio	42.12%	36.45%	20.86%	1.60	-20.59%
DJIA	17.95%	15.72%	18.86%	0.67	-18.76%
SPY	9.45%	8.31%	16.45%	0.32	-16.37%

Manager Commentary

- **April drawdown** came as a sharp Fed repricing and weak macro signals triggered a broad risk-off shift, pressuring our cyclical exposures while defensives only partially offset losses.
- **Performance recovery** followed as markets leaned into a policy-easing narrative, lifting high-quality cyclicals and financials that rebounded on improving demand visibility.
- **Our resilience was supported** by selective tech participation and a stable defensive core, which helped balance macro volatility and reduce exposure to speculative swings.

Looking Ahead

- **AI sentiment risk** remains a key swing factor as the market tests the limits of the AI bubble, benefiting our value-oriented DJIA tech exposure while still posing downside if narrative momentum fades.
- **A softening macro backdrop** may support defensive sectors such as retail, media, and utilities, yet margin pressure and consumer fatigue could curb the upside if labor conditions deteriorate further.
- **Potential Fed easing** offers a tailwind for high-quality cyclicals and financials, though a re-acceleration in inflation or policy hesitation could shift flows back toward defensives and dampen earnings visibility.

We stay moderately constructive but valuation-aware, keeping selective AI exposure and a quality bias while monitoring policy shifts and earnings momentum.



2 ASSET ALLOCATION & SECURITY ANALYSIS

MACRO OUTLOOK

The U.S. economy in late 2024 reflects a steady disinflation trend, with headline PCE running near 2.6% and CPI at 2.5%, though core measures remain above the Fed's 2% target, keeping policymakers cautious about the timing of rate cuts. Labor market conditions have continued to ease without weakening materially, as wage growth has slowed to the 3–4% range—reducing cost-push pressures and supporting the prospect of a non-inflationary expansion. Real GDP growth is moderating toward the mid-2% range, with consumption remaining resilient but no longer overheating, suggesting that aggregate demand is coming into better balance with supply. Long-term inflation expectations remain well anchored, and improved labor productivity has strengthened supply-side conditions, reinforcing the view that inflation is on a sustainable downward path. Overall, the macro environment points toward a soft-landing scenario in which the Fed can gradually normalize policy while preserving economic momentum.

SECTOR OUTLOOK

Technological Information

The U.S. IT sector entered late 2024 with strong numerical support for continued growth, anchored by a 19.2% YoY increase in global semiconductor sales in the first half of 2024 and full-year industry revenue forecasts of approximately USD 627–642 billion, implying roughly 19% annual growth. Enterprise technology investment remained a powerful demand engine, with double-digit expansion expected across software and IT services and public-cloud spending projected to rise more than 20% alongside U.S. enterprise IoT spending growth of roughly 13%. Long-term structural momentum was reinforced by sustained digitalization and AI-driven infrastructure needs, reflected in U.S. semiconductor sales rising from USD 71.1 billion in 2001 to USD 264.6 billion in 2023, a compound annual growth rate of about 6%. These quantitative indicators support an overweight stance with key drivers including (1) rapid semiconductor-industry expansion, (2) accelerating cloud and IT-services spending, and (3) structural digital-infrastructure and AI demand.

Financials

The U.S. financial sector is strengthening as core banking profitability improves, supported by rising NII and a 7 bps increase in system-wide NIM to approximately 3.23% as loan yields continue to outpace deposit costs. Revenue diversification is also gaining momentum, with many banks reporting more than 5% YoY growth in NOI driven by fee businesses and consumer services, which enhances earnings stability in a mixed macro environment. Large banks are expanding balance sheets, with the top fifty institutions adding roughly USD 377 billion in assets during the third quarter and benefiting from scale advantages that support resilience and competitiveness. These favorable trends suggest improving earnings visibility and operating leverage for Financials with key drivers including (1) margin expansion, (2) growth in diversified fee income, and (3) continued consolidation and asset growth.



Industrial

The U.S. financial sector remained fundamentally resilient despite softness in manufacturing indicators, supported by stable credit conditions and improving balance-sheet efficiency across banks. Large diversified institutions continued to benefit from steady loan demand in service-oriented parts of the economy, where activity held up better than manufacturing, helping sustain fee generation and overall earnings momentum. Sector fundamentals were reinforced by disciplined cost control and healthy capital positions, which position banks to capitalize on any eventual easing in monetary conditions. These favorable dynamics support an overweight stance with key drivers including (1) stable credit quality, (2) diversified revenue linked to services strength, and (3) solid capital and liquidity across major financial institutions.

Consumer Discretionary, Health Care, and Consumer Staples

Consumer Discretionary showed a mixed but improving backdrop, with September 2024 retail sales rising 0.4% and core sales up 0.7%, suggesting pockets of resilient demand that support higher portfolio weighting despite margin pressure and recent EPS downgrades. Health Care continued to offer steady, non-cyclical earnings with the sector up roughly 3% year-to-date by Q3 2024 and trading near 17 times forward earnings, aided by ongoing R&D investment and renewed M&A activity among large-cap biopharma. Consumer Staples remained operationally sound, though more defensive in profile, with year-to-date returns of approximately 5–6% supported by 3–5% revenue growth from pricing even as volumes faced ongoing down-trading pressure. These conditions support maintaining a relatively higher allocation to Consumer Discretionary, alongside balanced exposure to Health Care and Consumer Staples, with key drivers including (1) selective discretionary demand resilience, (2) stable health-care utilization, and (3) pricing power and margin management within staples.

Communication Services, Energy, and Materials

Communication Services entered late 2024 as one of the market's strongest performers, with sector indices up nearly 39% for the year, yet this momentum came with stretched valuations and growing sensitivity to regulatory and advertising-cycle risks. Energy fundamentals were supported by disciplined global production and stable-to-firm crude prices, but equity performance lagged as capital rotated toward growth sectors and as oil-price volatility and long-term decarbonization pressures created wide dispersion in earnings expectations. Materials faced the weakest backdrop of the three, with only about 47% of companies beating revenue estimates and sector-wide negative YoY revenue growth tied to soft industrial activity, excess inventories, and cautious capital spending across mining and chemicals. Despite these challenges, selective opportunities remained in high-cash-flow energy producers, premium communications platforms benefiting from digital-consumption trends, and specialty materials tied to infrastructure or energy-transition demand. These conditions justify underweight exposure while focusing on firms with (1) resilient demand drivers, (2) stable free cash flow despite macro volatility, and (3) clear strategic positioning to navigate regulatory, commodity, and growth-cycle risks.

ASSET ALLOCATION

Portfolio selection

As of 1 Oct 2024, the portfolio maintained a core U.S. equity allocation anchored to the DJIA universe, with selective sector tilts reflecting a forward-looking macro view. Allocation favored companies with earnings resilience, pricing power, and strong balance sheets under a "higher-for-longer" rate environment. The fund overweighted quality compounders in Technology, Healthcare, and Consumer Staples while underweighting cyclical and rate-sensitive sectors (Industrials, Materials, Energy) to manage downside asymmetry. Sector exposures were implemented through stock-level weights to preserve diversification. Security selection followed a quality-valuation framework emphasizing

- **Fundamentals:** high and stable ROIC/ROE, strong cash flows, prudent leverage, and consistent capital returns.
- **Valuation:** preference for reasonable forward multiples relative to sustainable growth and cash-flow durability.
- **Risk control:** liquidity screens and exclusion of names with unbalanced event risk.

Weights were derived using a robust mean-variance optimization with shrinkage to reduce estimation error, bounded by practical limits on position size and sector balance. Rebalancing was triggered by >5% drift or fundamental changes, ensuring alignment with the intended style and risk budget while minimizing turnover.

Portfolio optimization

To form our 2025 return forecasts, we applied the Black–Litterman (BL) model with Absolute Views. This framework starts from market-implied equilibrium returns and then adjusts them using our stock-specific views developed from fundamental work and sell-side analyst target prices and recommendations (e.g., JPMorgan, Barclays, Wells Fargo, etc.). By blending judgement with observed market risk information, BL produces expected returns that are more stable and economically reasonable than estimates based purely on historical averages.

Once the BL posterior expected returns were obtained, we performed portfolio optimization to maximize the expected Sharpe Ratio, under these implementation constraints:

- (1) Sector totals must match the allocations set in Section II.1 (Portfolio Selection).
- (2) Single-name exposure is limited so that each stock weight cannot exceed 10%.
- (3) The portfolio remains long-only and fully invested, meaning weights sum to 1 and no short positions (negative weights) are allowed.

FIGURE 4 | Analyst report with Target Price provided by Wells Fargo

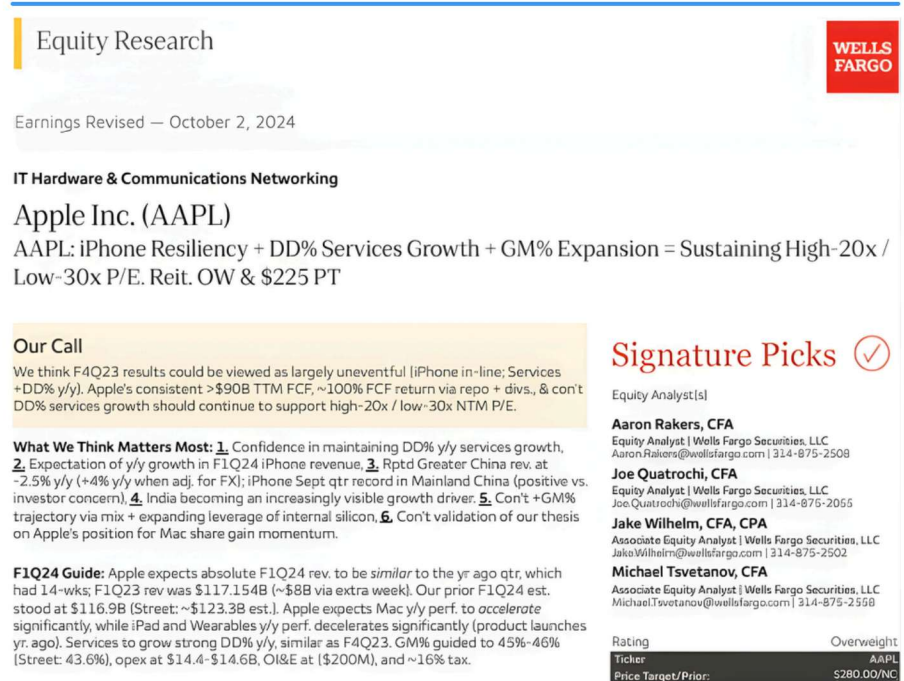


TABLE 2 | Integration of prior market equilibrium and subjective views into final allocations

Black–Litterman–Derived Optimal Security Weights

Sector	Sector Weight (%)	Ticker	Prior (%)	Posterior (%)	Shares (01/10/2024)	BL Optimal Weight (%)
Information Technology	21.8	INTC	11.3	40.92	355,254	16.7
		IBM	9.03	17.19	13,486	5.8
		CSCO	9.91	17.5	49,854	5.1
Financials	27.8	GS	10.39	27.54	11,117	10.7
		JPM	9.98	25.36	21,550	8.8
		AXP	10.94	26.82	13,620	7.3
Health Care	8.9	JNJ	7.74	3.85	28,494	8.9
Consumer Staples	8.9	WMT	7.76	11.77	55,582	8.9
Consumer Discretionary	8.9	MCD	8.56	4.4	15,027	8.9
Industrials	18.4	CAT	9.6	26.06	10,666	8.2
		BA	11.71	31.09	16,920	5.1
Energy	5.6	CVX	9.61	21.22	19,939	5.6

TABLE 3 | Black-Litterman Portfolio Performance

Metric	Value
Return of Portfolio	31.06%
Standard Deviation of Portfolio	19.78%
Sharpe Ratio	1.41

BOX 1 | How the Active Fund Is Built



PORTFOLIO CONSTRUCTION & REBALANCING



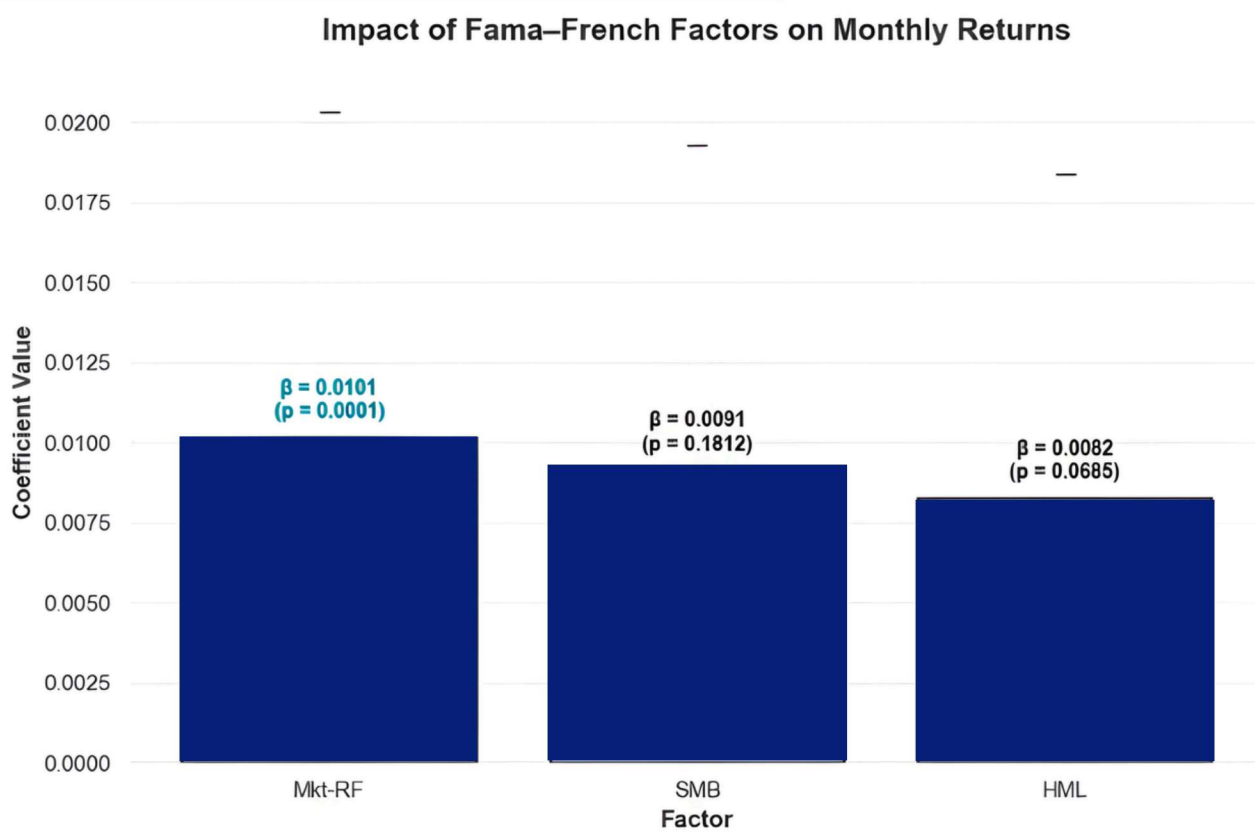
PORTFOLIO CONSTRUCTION

Fama–French Three-Factor Model

For our DJIA-based active portfolio, the factor regression shows that market risk (Mkt–RF) is the most influential driver of returns. This is consistent with a U.S. large-cap strategy where overall equity market conditions explain most of the performance variation, and our active decisions are expressed as tilts around a market core rather than a full decoupling from the benchmark.

The size factor (SMB) contribution is naturally limited because the DJIA universe is dominated by mega/large-cap companies, leaving minimal structural exposure to small-cap risk premia. Meanwhile, the value factor (HML) effect appears modest, indicating our portfolio is not a pure value bet; instead, it reflects a more quality-oriented large-cap profile embedded in our stock selection and BL-based weighting

FIGURE 5 | Impact of Fama–French Factors on Portfolio Returns



(*) Chart generated using the group's Python code.

Practically, we use Fama–French mainly as a style/risk diagnostic. For forward-looking construction, we rely on Black–Litterman Absolute Views to form expected returns and then optimize weights under our constraints (sector budgets, long-only, single-name limits).

Expected Risk Measurement

We measure forward-looking portfolio risk using a Monte Carlo simulation based on multivariate normal sampling. This approach is suitable for an equity portfolio because it **(i)** preserves the cross-asset dependence structure through the estimated covariance matrix and **(ii)** generates a full distribution of potential future portfolio values, rather than a single-point risk estimate.

FIGURE 6 | Value-at-Risk (VaR) Analysis of Ending Portfolio Value

Estimated probability density function with 95% and 99% VaR thresholds highlighted



(*) Chart generated using the group's Python code.

Method.

Starting from current prices (01/10/2024), we estimate each stock's return parameters and the covariance matrix. We then simulate 10,000 one-period return vectors from the multivariate normal distribution. To better align simulated returns with observed return shapes, we apply a %ile-to-%ile mapping (transforming simulated ranks to the empirical return distribution) and smooth the empirical distribution using kernel density estimation. Each simulated return vector is converted into new stock prices, which are aggregated using portfolio weights to obtain a simulated ending portfolio value.

REBALANCING

Framework

The Active Fund uses a drift-based rebalancing rule with a $\pm 5\%$ tolerance band around Black-Litterman optimized target weights. A full rebalance is triggered only if any holding's portfolio weight breaches this band.

Methodology

- **Initial capital:** USD 50,000,000
- **Drift threshold:** 0.05 ($\pm 5\%$)
- **Universe/targets:** 12 DJIA constituents, Black-Litterman optimized weights
- **Process:** Portfolio weights are monitored continuously; when a breach occurs, holdings are resized to restore target weights.

Backtest Results (01 Oct 2024 – 20 Nov 2025)

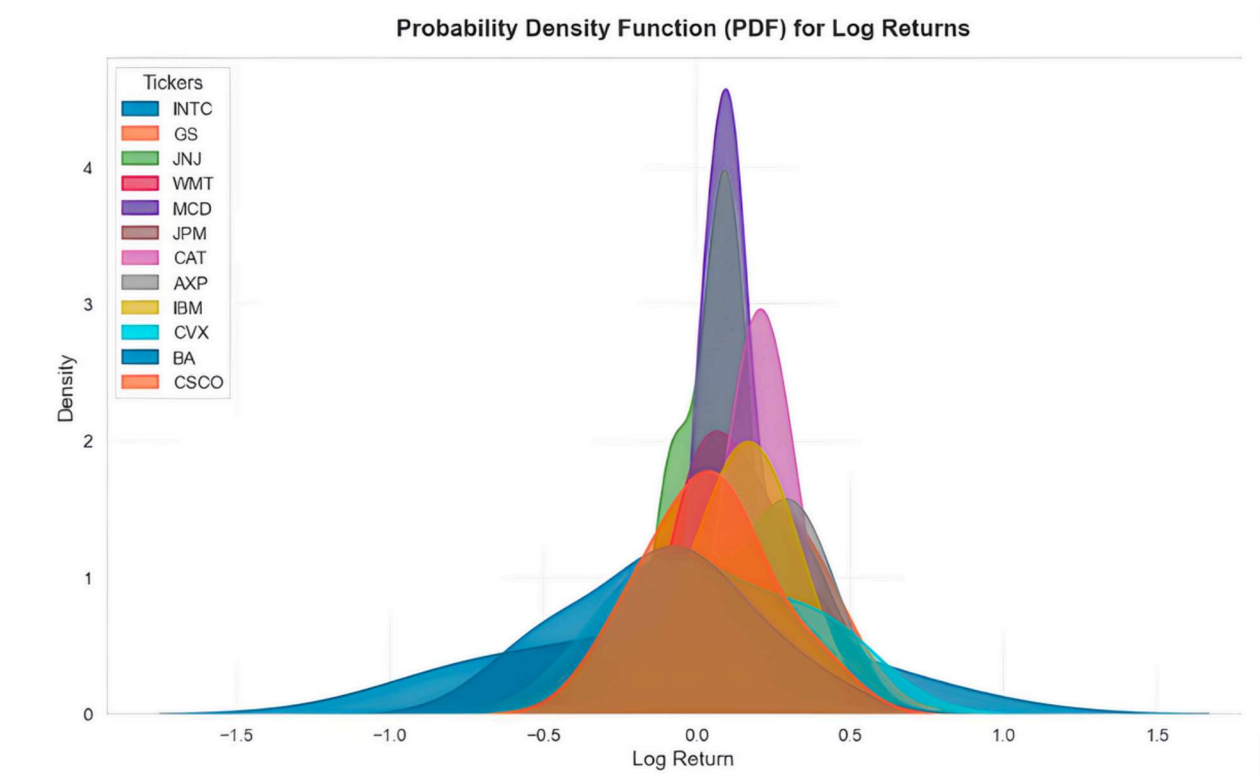
- **Rebalance events:** 0 (weights remained within the $\pm 5\%$ band throughout; portfolio stayed naturally balanced due to low cross-asset volatility and stable sector performance)
- **Final portfolio value:** USD 68,695,680.97
- **Total return:** +42.12%

Commentary

No rebalancing was triggered during the period, indicating that portfolio weight drift remained within the $\pm 5\%$ band. This outcome implies limited cross-sectional dispersion among holdings and that the initial optimized allocation was stable, reducing turnover and transaction costs while maintaining alignment with the Fund's intended exposures and risk profile.

FIGURE 7 | Probability Density Functions of Log Returns

Distribution of log return behavior across selected portfolio tickers



Source: Team Analysis

VaR extraction

The 10,000 ending values are sorted from lowest to highest to compute Value-at-Risk (VaR). The 99% VaR corresponds to the 1st %ile (extreme downside), while the 95% VaR corresponds to the 5th %ile. These cutoffs quantify the fund's potential loss under adverse but plausible market moves and provide a consistent basis for comparing risk across our active and passive strategies.

Scenario analysis

To stress-test portfolio robustness beyond the base period, we conducted scenario analysis by replaying our actual portfolio weights over three historical market regimes: a Bullish regime, a Bearish regime, and a Stable regime. For each regime, we re-estimated return distributions (log returns), preserved dependence via the covariance structure, and computed the portfolio-level path of wealth from an initial USD 50 million. This allows us to compare how the same portfolio construction behaves under different market environments, highlighting downside sensitivity and regime dependence.

TABLE 4 | Scenario Analysis: Active vs. Passive Fund Performance (2024–2025)

Returns, ending wealth, and total performance under bullish, bearish, and stable market conditions

Scenario	Fund	2024 Return	2025 Return	Final Wealth (USD)	Total Return (2024–2025)
Bullish	Active Fund	18.94%	18.94%	70,733,755.10	41.45%
Bearish	Active Fund	–28.92%	–30.64%	24,651,706.88	–50.61%
Stable	Passive Fund	18.46%	17.35%	69,588,027.00	45.58%



4 PERFORMANCE & RISK EVALUATION

PERFORMANCE MEASUREMENT

Over the evaluation window (01/10/2024–20/11/2025), our active DJIA-based portfolio delivered strong absolute and risk-adjusted performance relative to its benchmarks. As shown in Figure X, the portfolio's cumulative value trend stayed above both the DJIA (core benchmark) and SPY (broad market proxy), indicating that the active tilts added value beyond passive exposure.

FIGURE 8 | Portfolio and Market Performance (Portfolio vs. DJIA vs. SPY)

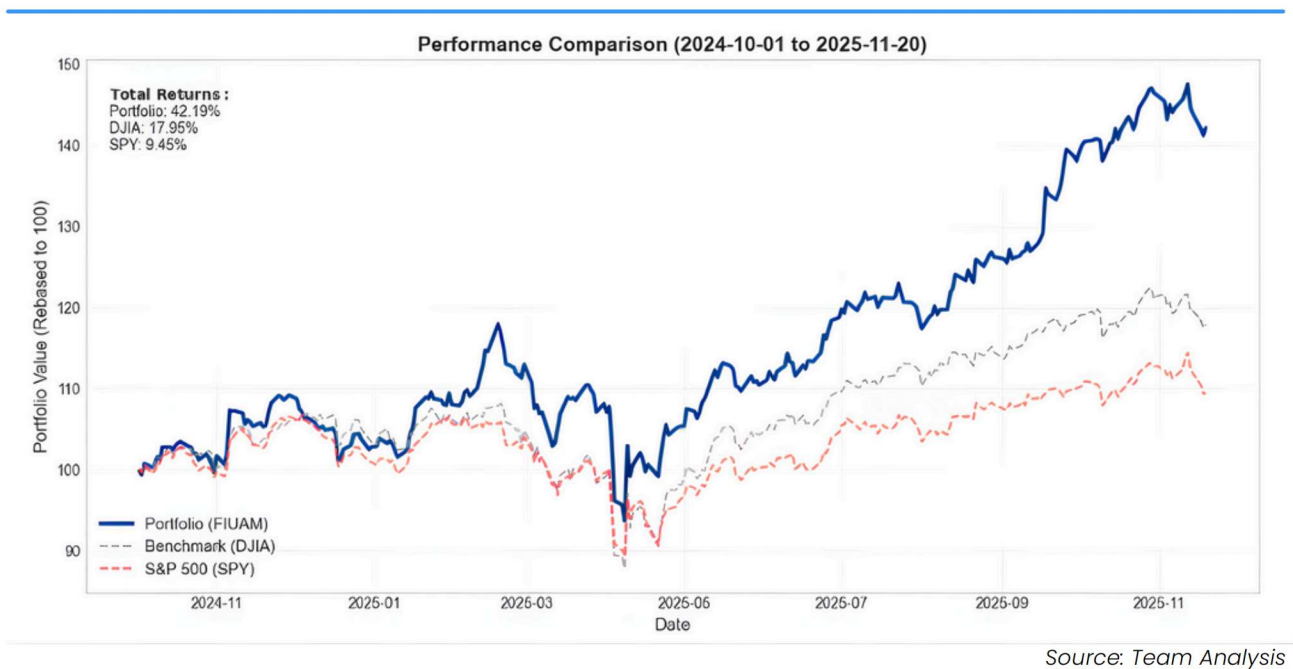


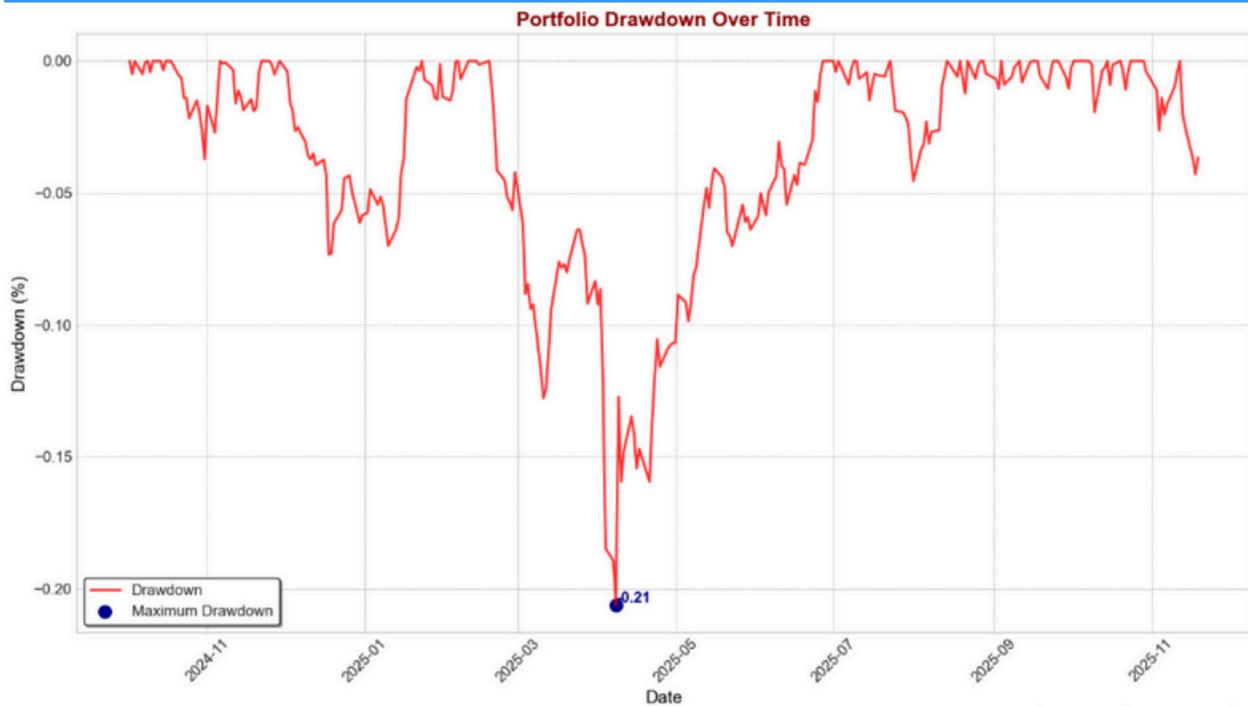
TABLE 5 | Comparative Performance Metrics: Portfolio vs. Benchmarks

Key return, risk, and drawdown statistics relative to the DJIA and SPY

Performance Metric	Portfolio	DJIA	SPY
Total Return	42.19%	17.95%	9.45%
Annualized Return (CAGR)	36.51%	15.72%	8.31%
Portfolio Volatility (Ann.)	20.86%	18.86%	16.45%
Sharpe Ratio	1.61	0.67	0.32
Maximum Drawdown	–20.61%	–18.76%	–16.37%

The annualized return (CAGR) of the portfolio (36.51%) was more than double the DJIA (15.72%) and far above SPY (8.31%), showing that the strategy delivered meaningful compounding outperformance over the full period rather than short-lived gains. This outperformance came with moderately higher annualized volatility (20.86% vs. 18.86% DJIA and 16.45% SPY), consistent with active tilts and name concentration. However, the portfolio's maximum drawdown (-20.61%) was only slightly deeper than DJIA (-18.76%) and SPY (-16.37%), suggesting that the portfolio took some additional downside risk, but it remained broadly controlled relative to the magnitude of return outperformance.

FIGURE 9 | Portfolio Drawdown Over Time



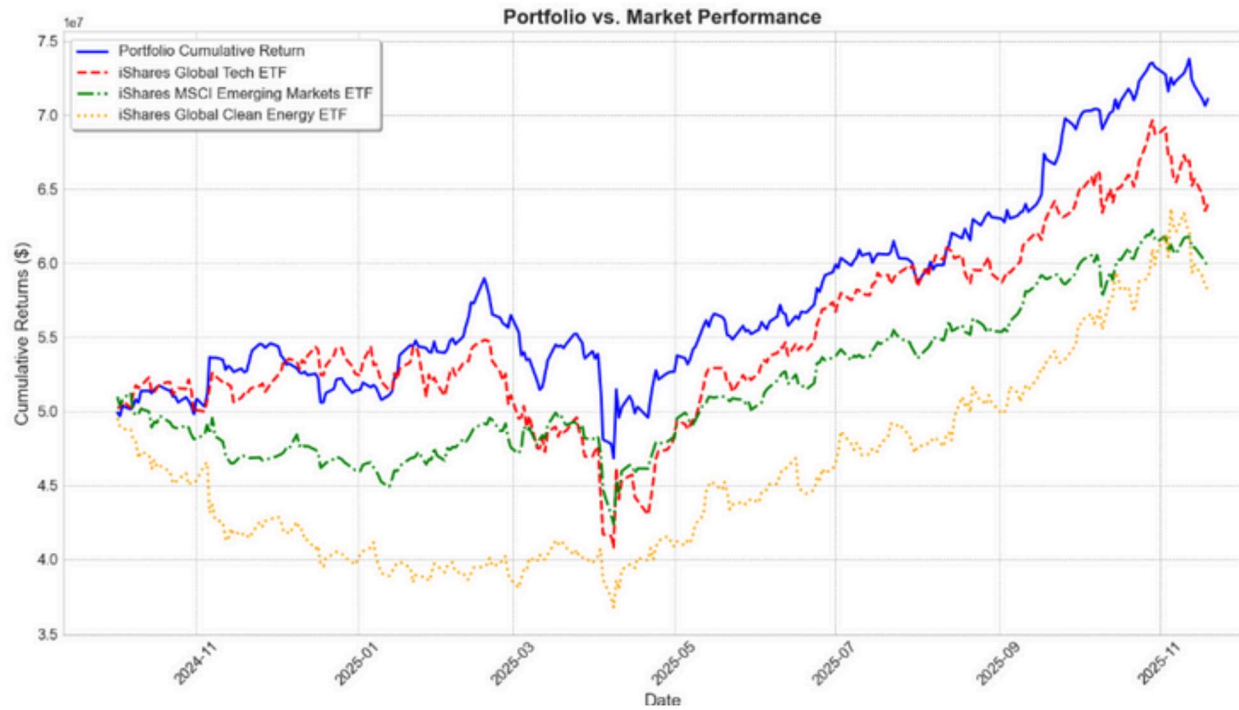
Source: Team Analysis

TABLE 6 | Systematic Exposure & Relative Performance Indicators

ETF	Beta	Correlation	Annualized Jensen's Alpha
iShares Global Tech ETF	0.57	0.72	0.15
iShares MSCI Emerging Markets ETF	0.69	0.59	0.17
iShares Global Clean Energy ETF	0.3	0.34	0.24

The portfolio's beta of 0.57 versus the iShares Global Tech ETF indicates materially lower sensitivity to the U.S. tech sector while still retaining meaningful exposure. In addition, the low correlation with the iShares Global Clean Energy ETF (0.30) supports the effectiveness of our diversification. Overall, the portfolio delivered strong performance with a solid risk-return trade-off. Going forward, we will prioritize downside risk control and reduce excess concentration in tech to improve long-term stability.

FIGURE 10 | Portfolio vs. Market Performance



Source: Team Analysis

TABLE 7 | Security-Level Performance Attribution

Ticker	Cumulative Return (%)	Weighted Return (%)
INTC	54.74	2.18
GS	63.62	3.05
JPM	50.5	2.46
JNJ	29.06	1.12
WMT	25	0.94
MCD	2.25	0.08
CAT	43.6	1.87
AXP	30.03	1.2
IBM	36	1.41
CVX	7.12	0.33
CSCO	53.63	2.09



PASSIVE FUND



EXECUTIVE SUMMARY

FUND DESCRIPTION

Our Investment Philosophy

The Trust seeks to provide investment results that, before expenses, correspond generally to the price and yield performance of the Dow Jones Industrial Average (the "DJIA").

Our Investment Objective

- **Benchmark:** Dow Jones Industrial Average (DJIA)
- **Annualized Return Target:** Match DJIA (tracking error < 5%)
- **Expected Total Return:** Match benchmark performance
- **Volatility Target:** Within $\pm 2\%$ of DJIA volatility
- **Sharpe Ratio Target:** ≥ 0.35
- **Maximum Drawdown Limit:** < 20%
- **Beta (β)** ≈ 1.0 (index-tracking)
- **Futures Usage:** Cash equitization / exposure alignment

Fund Information

Inception Date	1 October, 2024
AUM	\$ 50 million
Benchmark Index	Dow Jones Industrial Average

PERFORMANCE SUMMARY

FIGURE 11 | Performance Comparison: Passive Fund vs. DJIA Benchmark

Normalized daily performance (Base = 1.0), illustrating tracking accuracy of the passive portfolio

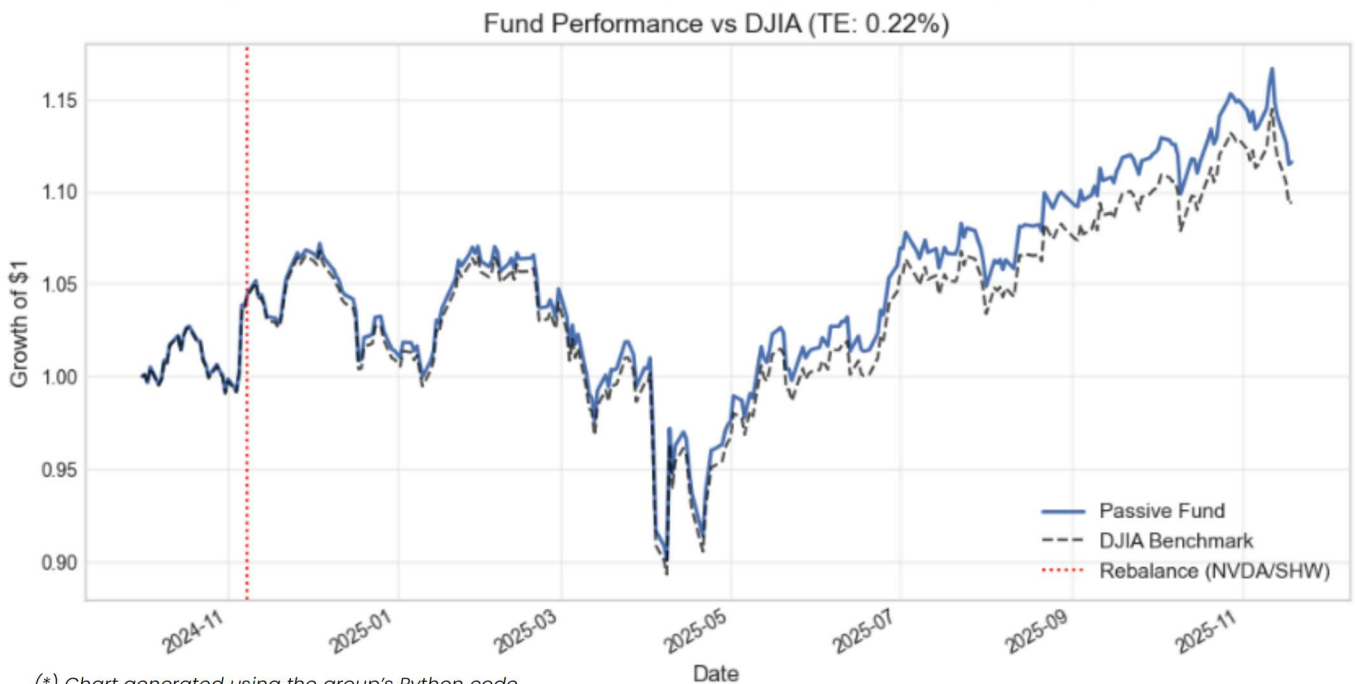


TABLE 8 | A concise snapshot of the fund's return, risk, and tracking accuracy relative to the DJIA

Note: Performance results are based on the unhedged passive portfolio prior to futures overlay.

Ending Value	Total Return	Volatility	Sharpe Ratio	Tracking Error
\$55,795,043.01	11.59%	16.45%	0.43%	0.22% (Target < 5%)

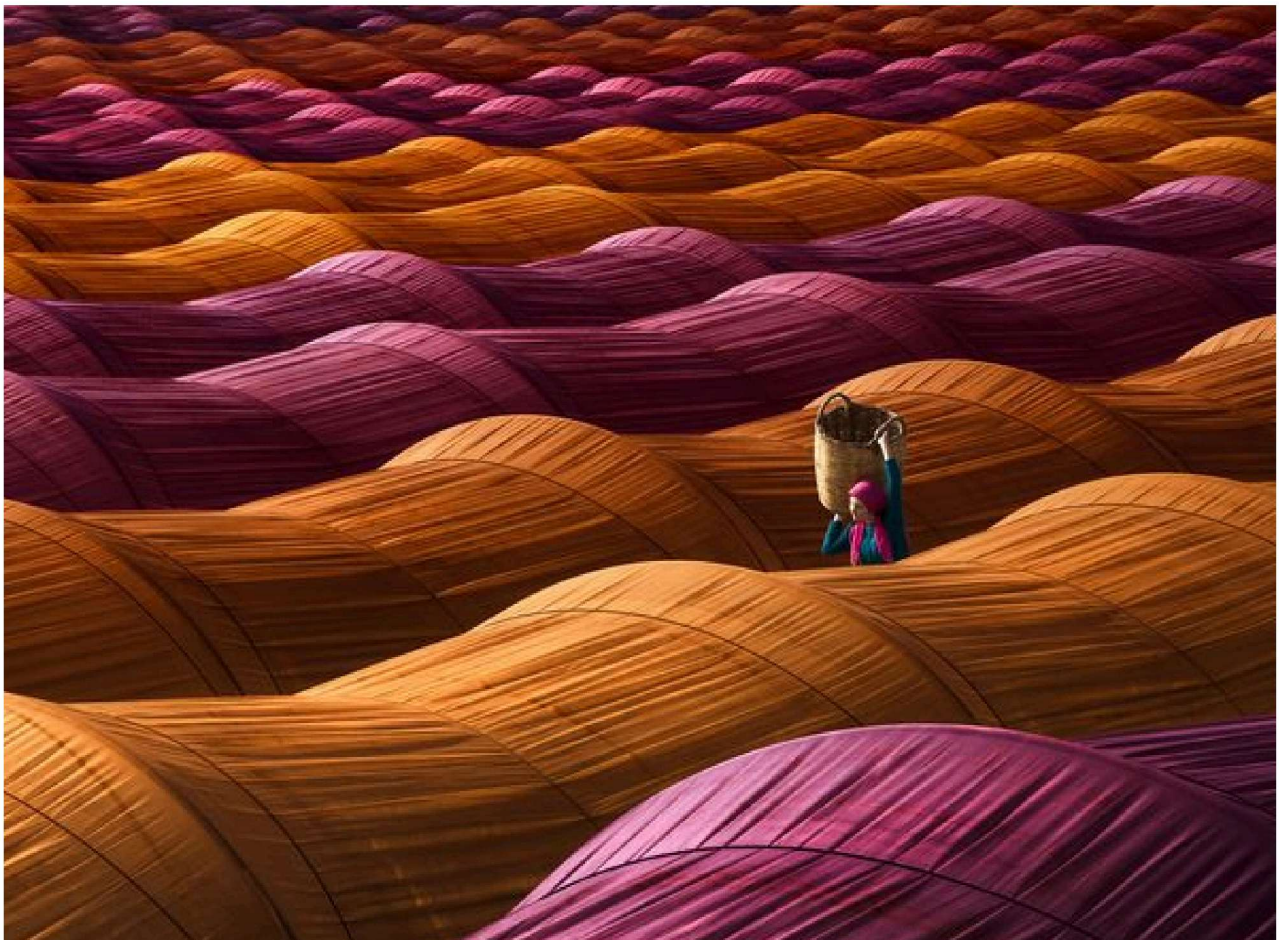
Manager Commentary

- **Sharp April drawdown:** The DJIA fell sharply in early April 2025, driven by a sudden U.S. tariff announcement that triggered fears of a global trade slowdown, risk-off sentiment, and a broad equity sell-off—marking one of the index’s worst single-day declines in years.
- **Resilient mid-year recovery:** By mid-2025, investor confidence improved as trade tensions eased and large-cap earnings showed greater-than-expected resilience, helping to support a steady rebound in blue-chip valuations.
- **Sustained uptrend into Q3:** From the April low to October 3, the DJIA posted a strong recovery, aided by improved macro stability, a rotation back into diversified large-cap sectors, and continued interest in high-quality equities amid moderating volatility.

Looking Ahead

- **Macro tailwinds may support value-led rebound:** With inflation softening and Fed rate cuts increasingly priced in for mid-2026, DJIA’s value and dividend-tilted structure may benefit as equity markets reprice toward cyclicals and cash-generative companies.
- **Earnings stability in core sectors:** Forecasts suggest continued earnings resilience among large-cap industrials, financials, and consumer staples—sectors where DJIA has broad exposure—offering potential downside protection if speculative tech momentum fades.
- **Volatility and election risk may shift flows to defensives:** As policy uncertainty and geopolitical tensions rise ahead of the 2026 U.S. elections, DJIA’s bias toward high-quality, stable cash-flow names could attract capital rotation away from more volatile, growth-dependent sectors.

While macro and policy uncertainty remain elevated, the DJIA’s diversified and value-oriented composition positions it well to navigate a shifting market regime—potentially benefiting from defensive rotation, rate normalization, and stable large-cap earnings.





PORTFOLIO CONSTRUCTION & REBALANCING

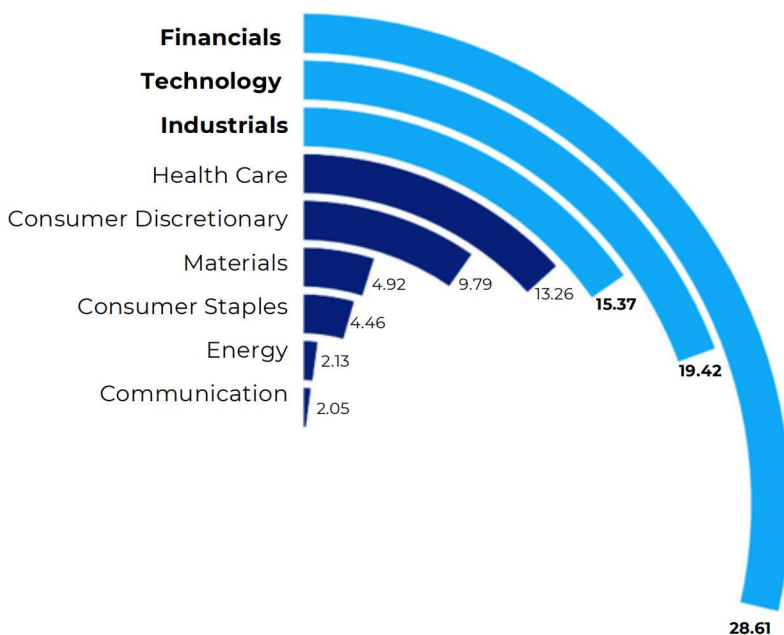
(*) To incorporate real-world market frictions while maintaining a tractable framework, we assume that at fund inception on 1 October 2024, 95 percent of the USD 50 million NAV is invested in DJIA constituents, accounting for execution delays, settlement frictions, and slippage, with the remaining 5 percent held temporarily in cash.

INVESTMENT APPROACH & REBALANCING POLICY

The Fund is managed as a passive long only portfolio designed to replicate the DJIA. Holdings are maintained according to each constituent's price weighted contribution to the index. Portfolio weights are allowed to drift between scheduled rebalancing points to limit turnover and reduce transaction costs. Rebalancing is event-triggered, occurring only when index composition changes or when corporate actions such as splits or spin-offs alter index calculations. Required adjustments add or remove constituents and realign security weights to preserve index accuracy. The objective is to maintain benchmark consistency while operating within a tracking error tolerance below 5 % per annum.

FIGURE 12 | Where the portfolio is tilted

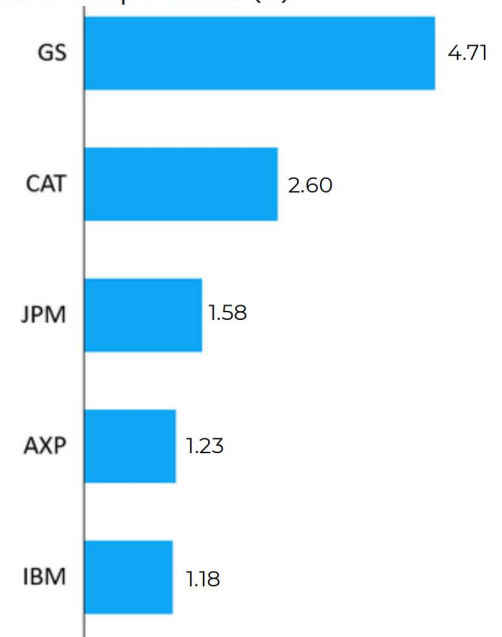
Price-Weighted Sector Allocation (%)



(**) Data derived from the group's Python dataset.

TABLE 9 | What drove the fund's gains

Return of top 5 tickets (%)



(***) Data derived from the group's Python dataset.

INDEX FUTURES HEDGING & CASH EQUITIZATION

Cash equitization refers to the use of index futures to transform non-invested cash into synthetic equity exposure, allowing the portfolio to maintain a return profile that closely mirrors the underlying benchmark. We employ this approach because it directly supports the Trust's mandate to deliver results that correspond to the price and yield performance of the Dow Jones Industrial Average, ensuring that short-term cash balances do not weaken alignment with the index. As the portfolio begins with a 5% cash position that would otherwise create measurable tracking error and dilute benchmark sensitivity, we establish a DJIA futures position—calculated under the theoretical framework below—to offset this drag and preserve full exposure.

BOX 2 | Illustration of Cash Equitization Hedge Ratio

To equitize idle cash using DJIA index futures, the number of contracts is determined using the standard hedge ratio formula:

$$\text{Number of Contracts} = \frac{\text{Cash Amount to Equitize}}{\text{Index Level} \times \text{Contract Multiplier}}$$

Rounding to the nearest whole number, equitizing the USD 2.5 million cash balance requires 6 long DJIA futures contracts to deploy the 5 percent cash position and preserve full index exposure. This adjustment reduces the estimated tracking error from 0.8226 percent to 0.0000 percent, demonstrating the effectiveness of futures in eliminating cash drag.

TABLE 10 | Hedged Passive Fund Performance (2024-10-01 to 2025-11-20)

Metric	Value
Total Return	11.37%
Volatility (Annualized)	16.45%
Sharpe Ratio (Annualized)	0.427358
Tracking Error vs DJIA	0.21%
Max Drawdown	−15.9842%
Ending NAV (Corrected)	\$55,740,933.47



PERFORMANCE & RISK EVALUATION

Rebalancing Update

On November 8, 2024, the DJIA replaced Intel with NVIDIA and Dow Inc. with Sherwin-Williams; therefore, triggering our rebalancing policy, we reconstituted and rebalanced our portfolio accordingly. [1]

Unhedged case

We achieved performance that closely matched the DJIA, reflected in a 1.47 % tracking error and a rolling beta near 1.0 that confirms our price-weighted replication held benchmark risk throughout the period. These forces allowed our fund to capture the benchmark's return dispersion with minimal deviation. Return attribution shows that GS, CAT, JPM, AXP, and IBM dominated gains, driven by stronger financial-sector earnings, ongoing industrial demand supporting CAT's backlog, and rotation into cash-generative enterprise tech benefiting IBM. When markets dislocated in April, we absorbed a 15.6 % drawdown versus the DJIA's 16.4 %, demonstrating that the portfolio maintained symmetric downside capture consistent with an unhedged passive design.

Hedged Case (Overlay Strategies)

By implementing a short DJIA futures overlay, we shifted the portfolio from full market participation to a beta-neutral stance, which reduced returns to 0.85 % but also compressed volatility to 1.47 % compared with the unhedged fund's 10.44 % return and 16.21 % volatility. The hedge proved effective during periods of market stress, as gains on the short futures position offset index losses and prevented the portfolio's drawdown path from tracking the DJIA's full decline. This offsetting effect flattened peak-to-trough losses and demonstrated how index futures can neutralize systematic risk without altering underlying holdings. Overall, the futures overlay provided a clear mechanism for capital preservation when market conditions turned sharply negative.

TABLE 11 | Holdings Decision Summary

This reflects the updated position weights after the rebalance.

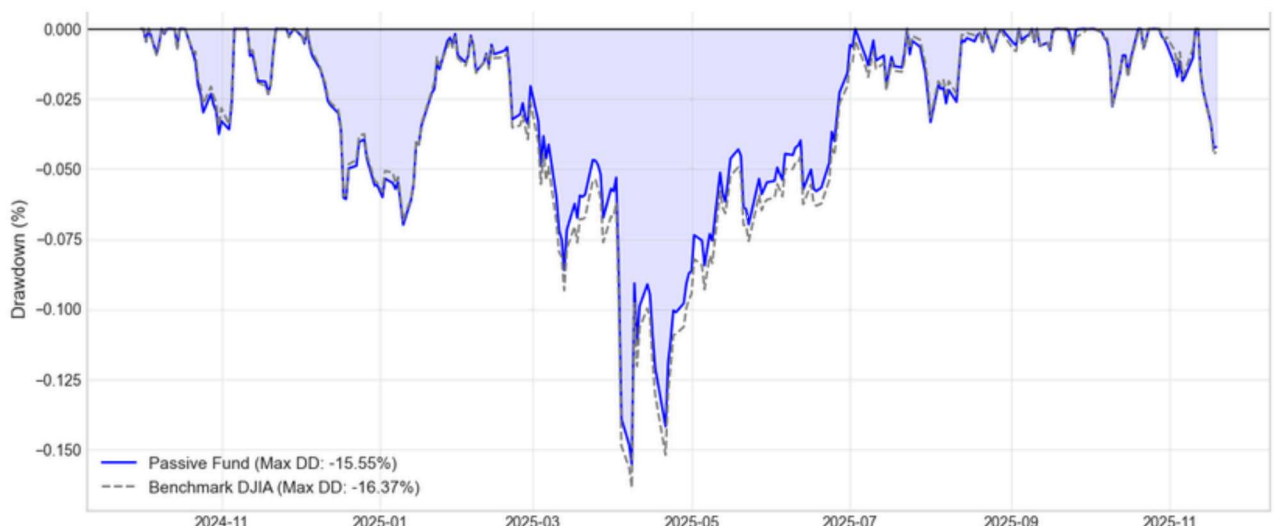
✓ NVDA: HELD (Value: \$1,403,231)

✗ INTC: REMOVED

✓ SHW: HELD (Value: \$2,531,155)

✗ DOW: REMOVED

FIGURE 13 | Historical Drawdown

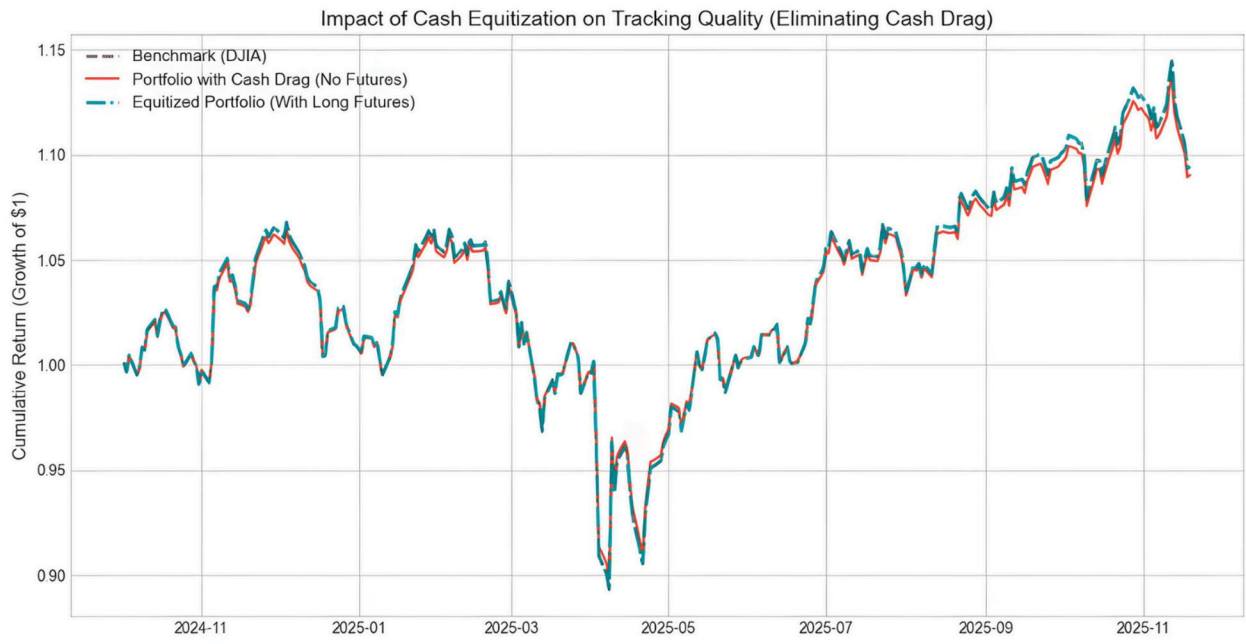


(*) Chart generated using the group's Python code.

[1] <https://www.withum.com/resources/dow-jones-industrial-average-changes-making-history/>

FIGURE 14 | Offsetting cash drag with futures

Cumulative-return trajectories plotted over time against the DJIA



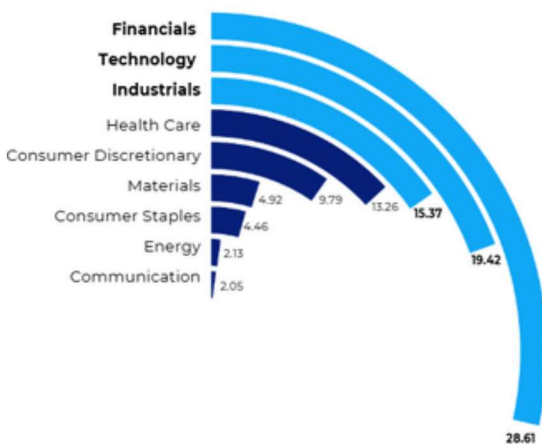
(*) Chart generated using the group's Python code.

Performance effects & Tracking error considerations

The hedge altered the relationship between the Fund and the DJIA by removing beta exposure. As a result, tracking error increased relative to a pure passive mandate because futures gains and losses diverged from benchmark return movements. When used for cash equitization, however, futures positions reduce tracking error by preventing unintended underexposure. Overall performance outcomes reflect the theoretical effect of futures overlays on passive portfolios, including **(1)** lower volatility, **(2)** reduced sensitivity to market drawdowns, and **(3)** divergence from benchmark returns when the objective shifts from replication to hedging.

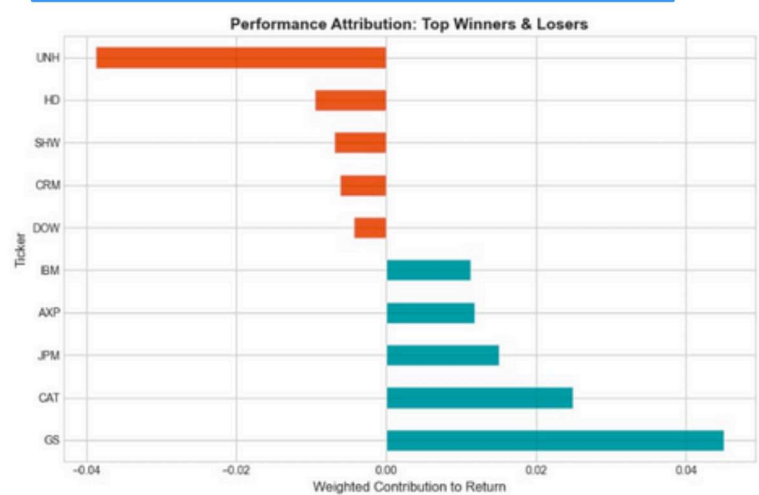
FIGURE 15 | Where the portfolio is tilted

Price-Weighted Sector Allocation (%)



(*) Data derived from the group's Python dataset.

FIGURE 16 | Performance Attribution: Winners & Losers



(**) Data derived from the group's Python dataset.



REFLECTION & LIMITATIONS



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REFLECTION

Active management proved effective in delivering superior risk-adjusted performance versus the passive benchmark. Over the evaluation period, the Active Fund generated a **Sharpe ratio of 1.60** (vs. **DJIA: 0.67**) and achieved an **annualized return (CAGR) of ~36.45%** (vs. **~15.72%**), while keeping risk within institutional bounds with **volatility < 20%** and a **maximum drawdown of ~-20.59%**. These results are consistent with the strategy's **quality-growth and earnings-resilience tilt**, supported by **Black-Litterman optimization** and a **±5% drift-based rebalancing** rule that maintained diversification without unnecessary turnover.

In contrast, the Passive Fund fulfilled its objective of disciplined index replication, delivering **tracking error < 5%** and **$\beta \approx 1.0$** , but—by design—produced a lower risk-adjusted outcome (**Sharpe: 0.43**) and a more modest total return (**~11.37%**) due to the absence of active tilts or alpha-seeking trades. Overall, the evidence supports the conclusion that **selective, process-driven active management** can add value when implemented within a **disciplined, risk-controlled framework**, combining fundamental screening, systematic optimization, and rules-based rebalancing to improve both absolute and risk-adjusted returns.

LIMITATIONS

We acknowledge that our modeling framework has theoretical limitations that may affect portfolio construction under certain market conditions. Key limitations include the following:

1. **Model dependence on prior assumptions**, because equilibrium returns and forward views embed structural estimation risk that can lead to allocation shifts when actual risk premia deviate from theoretical conditions.
2. **Instability of covariance structures**, since mean-variance optimization assumes stationary relationships among assets even though correlations and volatilities can reprice abruptly across market regimes.
3. **Simplifying distributional assumptions**, where reliance on approximately normal return behavior may understate tail dependencies and nonlinear risks that emerge during periods of market stress.
4. **Abstraction from implementation frictions**, because the theoretical portfolio ignores transaction costs and liquidity constraints, which can introduce divergence between optimized weights and feasible real-world allocations.

We recognize that our passive approach has inherent constraints, and we will continue to refine our tracking process. The main limitations include the following:

1. **Futures basis risk cannot be fully modeled**, because available data do not permit reliable measurement of pricing differentials between the index and exchange-traded contracts.
2. **Exposure precision is inherently limited**, since contract-level rounding restricts exact notional replication and slippage effects cannot be quantified without more granular execution data.
3. **Transaction costs remain unmodeled**, as roll-related costs and bid-ask dynamics in the futures market cannot be estimated with sufficient accuracy using current inputs.



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